



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

22AI503/22AM503 MACHINE LEARNING

UNIT II – DATA PREPROCESSING

Data Reduction

1. Introduction

1.1 Definition

Data reduction obtains a reduced representation of a dataset that is much smaller in volume yet produces the same or nearly the same analytical results. It addresses the challenges of high dimensionality, large sample sizes, and storage and computation bottlenecks.

Definition:

Goal: Reduce data volume (size, dimensionality, or numerosity) while preserving information quality.

Four strategies: Dimensionality Reduction, Numerosity Reduction, Data Compression, Feature Subset Selection.

Benefit: Faster training, less memory, reduced overfitting risk, sometimes better model accuracy.

1.2 Importance in Machine Learning

The curse of dimensionality states that as feature count d increases, the data volume needed for representative sampling grows exponentially. High dimensionality leads to sparse feature spaces, distance metrics losing discriminative power, and increased overfitting. Data reduction directly counteracts these problems. For datasets with hundreds to thousands of features, reduction is not optional — it is necessary.

1.3 Role in Data Preprocessing Pipeline

Data reduction is performed after cleaning and integration. It may be applied as feature selection (choosing which features to keep), dimensionality reduction (transforming to fewer dimensions via PCA or SVD), or numerosity reduction (sampling, clustering, parametric models). It feeds directly into the modelling step.

2. Learning Outcomes

Understand

- The four categories: dimensionality, numerosity, compression, and feature selection
- PCA algorithm: covariance matrix, eigendecomposition, explained variance, scree plot
- Sampling strategies: simple random, stratified, cluster, systematic

Apply

- Implement PCA using scikit-learn and interpret explained variance ratio
- Apply filter, wrapper, and embedded feature selection methods
- Use LDA for supervised dimensionality reduction

Analyze

- Compare PCA vs LDA: unsupervised vs supervised, variance vs class separability
- Evaluate the trade-off between compression ratio and information loss
- Select appropriate sampling strategy for imbalanced datasets

3. Detailed Technical Concepts

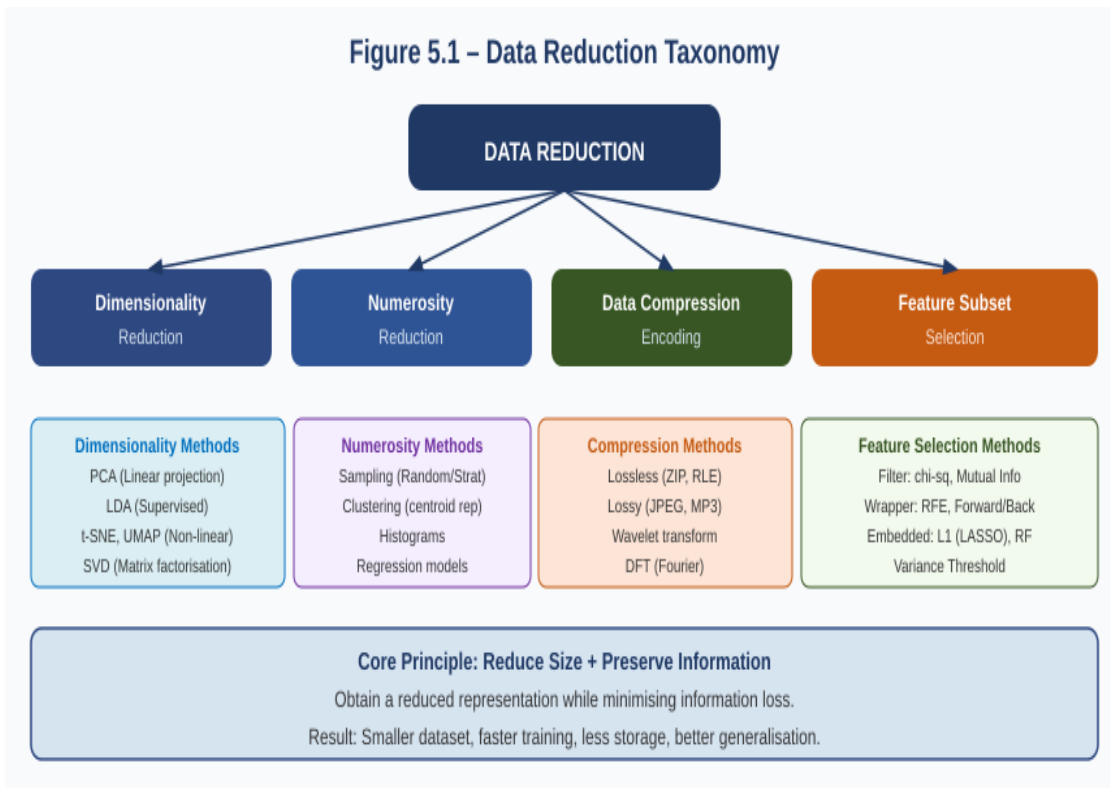


Figure 5.1 – Data Reduction Taxonomy: Four Main Categories

3.1 Dimensionality Reduction

3.1.1 Principal Component Analysis (PCA)

PCA finds the linear transformation of the original d -dimensional space to a k -dimensional subspace ($k < d$) that retains maximum variance. It is the most widely used unsupervised dimensionality reduction technique.

Algorithm Steps:

- Step 1: Standardise data — Z-score: $X_std = (X - \text{mean}) / \text{std}$ per feature
- Step 2: Compute covariance matrix: $C = (1/(n-1)) \times X_std^T \times X_std$ [$d \times d$ matrix]
- Step 3: Compute eigenvalues and eigenvectors of C
- Step 4: Sort by eigenvalue descending. Each eigenvector = one principal component
- Step 5: Select top k PCs where cumulative explained variance is at least 90-95%
- Step 6: Project data: $X_pca = X_std \times V_k$ where $V_k = [v_1, v_2, \dots, v_k]$

Explained Variance Ratio: $EVR_i = \text{eigenvalue}_i / \text{sum}(\text{all eigenvalues})$. Scree plot shows EVR per PC. Choose k at the elbow or where cumulative EVR reaches 90%.

sklearn usage: `PCA(n_components=0.95)` retains components explaining 95% of variance automatically.

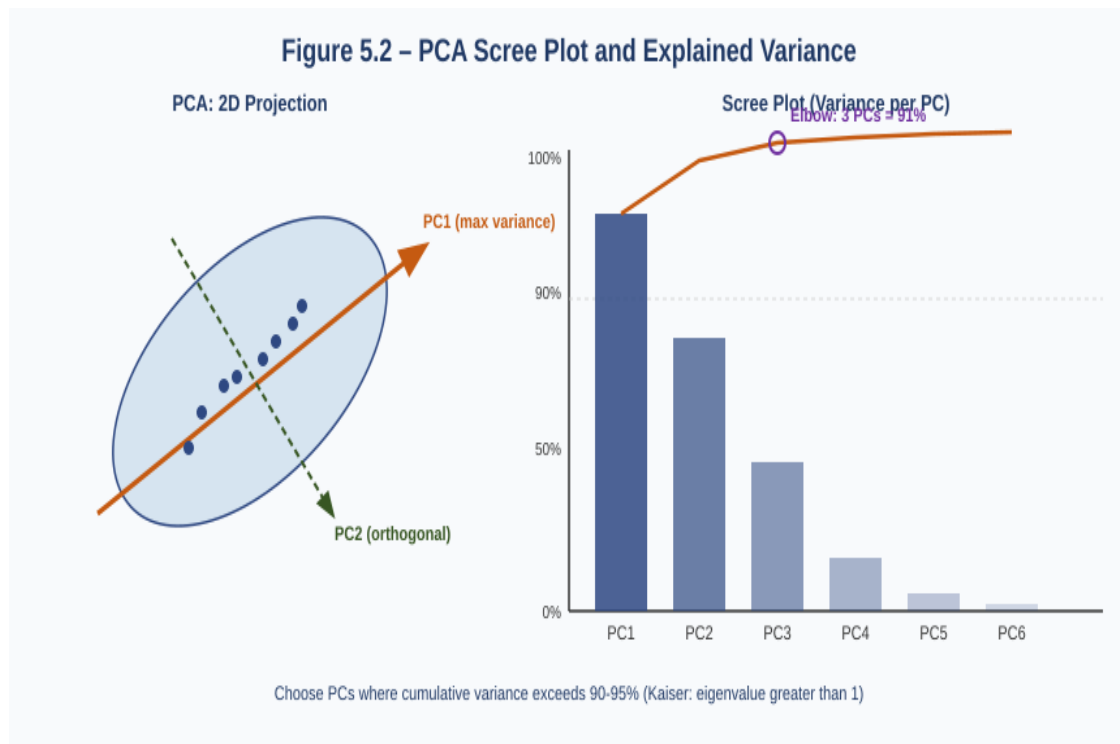


Figure 5.2 – PCA 2D Projection and Scree Plot for Choosing Number of Components

3.1.2 Singular Value Decomposition (SVD)

SVD decomposes any matrix A as $A = U \times S \times V^T$, where U holds left singular vectors, S holds singular values on the diagonal, and V holds right singular vectors. Truncated SVD keeps only the top k singular values and vectors. It is used in recommendation systems (matrix factorisation), NLP (Latent Semantic Analysis), and image compression. PCA on zero-mean data is mathematically equivalent to SVD of the data matrix.

3.1.3 Linear Discriminant Analysis (LDA) — Supervised Reduction

LDA finds linear combinations of features that best separate multiple classes. Unlike PCA which maximises total variance, LDA maximises class separability as the ratio of between-class scatter to within-class scatter. Maximum number of components: $k \leq \min(d, C-1)$ where C is the number of classes. LDA is both a dimensionality reduction technique and a classifier.

PCA vs LDA: PCA is unsupervised (no class label needed) and maximises total variance. LDA is supervised and maximises class discrimination. LDA typically outperforms PCA on classification tasks when class labels are reliable.

3.2 Numerosity Reduction

Numerosity reduction reduces the number of data objects (rows/samples) rather than the number of attributes (columns). This reduces training time proportionally to the reduction ratio.

3.2.1 Parametric Methods

Fit a statistical model to the data and store only the model parameters. Example: fit a linear regression $y = w_0 + w_1x_1 + w_2x_2$ and store only w_0, w_1, w_2 instead of all n data points. Loss-y when data does not fit the assumed model form.

3.2.2 Non-Parametric Sampling Methods

Sampling Type	Description	When to Use	Limitation
Simple Random Sampling (SRS)	Each record has equal probability of selection. With/without replacement.	Homogeneous datasets	May under-represent rare classes
Stratified Sampling	Sample proportionally from each class/stratum separately.	Imbalanced class datasets	Requires class label at sampling time
Cluster Sampling	Divide into clusters; randomly select entire clusters.	Geographically distributed data	High variance if clusters are heterogeneous
Systematic Sampling	Select every k-th record from sorted/ordered list.	Ordered datasets (time series)	Periodic bias if data has cyclic patterns
Bootstrap Sampling	Sample n records with replacement from n-record dataset. ~63.2% unique.	Model evaluation (bagging, RF)	Not suitable for rare event detection

3.2.3 Clustering-Based Reduction

Run k-Means or hierarchical clustering. Replace each cluster with its centroid (or medoid). The dataset size reduces from n records to k cluster representatives. This is lossy but dramatically reduces size for large datasets.

3.2.4 Histogram-Based Reduction

Partition attribute values into buckets and store the bucket boundaries and frequencies instead of raw values. Types: equal-width (same interval per bucket), equal-frequency (same count per bucket), V-optimal (minimises variance within buckets).

3.3 Data Compression

- Lossless compression: Reconstructs original data exactly. Examples: Run-Length Encoding (RLE), Huffman coding, ZIP/GZIP. Used for databases where exact reconstruction is required.
- Lossy compression: Approximates original data. Examples: JPEG image compression (DCT-based), MP3 audio (perceptual encoding). Acceptable when small information loss is tolerable.
- Wavelet Transform: Decomposes signal into frequency components. Truncating high-frequency coefficients gives lossy compression while preserving low-frequency structure. Used in seismic data, image/audio compression, and time-series reduction.
- Discrete Fourier Transform (DFT): Represents time-series data in frequency domain. Keep low-frequency components, discard high-frequency noise. Used in signal processing and temporal data mining.

3.4 Feature Subset Selection

Feature subset selection (also called feature selection) identifies the most informative subset of attributes from the original feature set and discards the rest. Unlike PCA, it keeps original features intact — improving interpretability.

Figure 5.3 – Feature Selection: Filter vs Wrapper vs Embedded

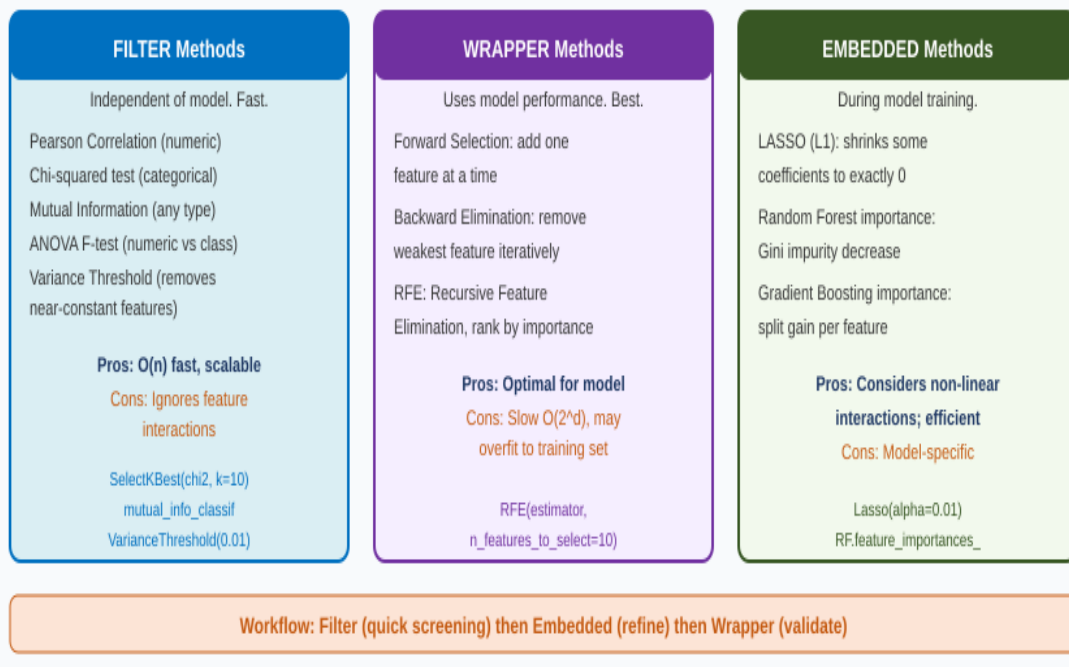


Figure 5.3 – Feature Selection Methods: Filter, Wrapper, and Embedded

3.4.1 Filter Methods

Evaluate each feature independently using a statistical score, then select top-k features above a threshold. Common scores: Pearson correlation (numeric), Chi-squared (categorical vs target), Mutual Information (any type), ANOVA F-test (numeric vs categorical class), Variance Threshold (remove near-constant features).

sklearn: `SelectKBest(score_func=chi2, k=10).fit_transform(X_cat, y)` selects 10 best categorical features by chi-squared score.

3.4.2 Wrapper Methods

Evaluate feature subsets by training a model and measuring cross-validated accuracy. Computationally expensive but finds the optimal subset for a specific model.

- Forward Selection: Start with empty set. Greedily add the feature that gives the best model improvement until no improvement or target count reached.
- Backward Elimination: Start with all features. Greedily remove the least useful feature until performance drops or target count reached.
- Recursive Feature Elimination (RFE): Train model, rank features by importance, remove weakest, repeat. sklearn: `RFE(estimator=LogisticRegression(), n_features_to_select=10)`.

3.4.3 Embedded Methods

Feature selection occurs during model training as part of the learning algorithm. Combines benefits of filter (speed) and wrapper (model-aware) methods.

- LASSO (L1 Regularization): Adds L1 penalty to regression loss: $\text{Loss} + \alpha \times \sum(|w_i|)$. This shrinks some coefficients to exactly zero, effectively removing those features. sklearn: `Lasso(alpha=0.01)`.

- Random Forest Feature Importance: Measures the average reduction in Gini impurity (or permutation importance) caused by each feature across all trees. Features with low importance are pruned.
- Gradient Boosting Importance: Measures the average gain in split quality for each feature across all boosting iterations. XGBoost: `model.feature_importances_`.

4. Data Reduction Workflow

Step	Action	Method	Output
1	Profile features	Variance threshold, correlation matrix	List of low-variance and highly correlated features
2	Remove redundant features	Drop features with $ r > 0.95$ or $\text{variance} < 0.01$	Reduced feature set
3	Filter selection	SelectKBest with <code>mutual_info</code> or <code>chi2</code>	Top-k informative features
4	Apply PCA (optional)	PCA(<code>n_components=0.95</code>)	k principal components explaining 95% variance
5	Validate	Cross-validated accuracy before vs after	Confirm accuracy maintained or improved
6	Sample if needed	Stratified sampling for imbalanced data	Balanced, smaller training set

5. Industry Tools and Technologies

5.1 Commercial Tools

Tool	Vendor	Reduction Features
SAS Enterprise Miner	SAS	PCA, variable selection node, partial least squares
IBM SPSS Modeler	IBM	PCA, feature selection, sampling nodes
DataRobot	DataRobot	AutoML with automated feature selection and importance ranking
H2O AutoML	H2O.ai	Built-in feature importance, LASSO, and RF selection
Google Vertex AI	Google	Auto feature engineering, BigQuery ML feature importance

5.2 Open-Source Tools

- scikit-learn: PCA, TruncatedSVD, LinearDiscriminantAnalysis, SelectKBest, RFE, VarianceThreshold, Lasso
- sklearn.decomposition: FastICA, NMF (Non-negative Matrix Factorisation), FactorAnalysis
- umap-learn: UMAP for non-linear dimensionality reduction
- mlxtend: SequentialFeatureSelector (forward/backward wrapper)
- boruta: Boruta algorithm — all-relevant feature selection using RF
- featuretools: Automated deep feature synthesis

6. Real-World Applications

Domain	Data Volume Challenge	Reduction Method	Result
Genomics	20,000 gene expression features per patient	PCA to 50 PCs + LASSO	Identified 12 key genes for cancer prediction
Computer Vision	784 pixel features per MNIST image	PCA to 154 PCs (95% variance)	kNN accuracy: 97.1% with 5× speed-up
Text Mining (NLP)	10,000-dim TF-IDF bag-of-words matrix	TruncatedSVD (LSA) to 300 dims	Topic clustering improved, noise reduced
Credit Scoring	200 raw financial features	RF importance + RFE to 25 feats	Same AUC (0.84), 8× faster inference
IoT Sensor Data	1M records/day from 50 sensors	Stratified sampling to 100K/day	98% statistical accuracy preserved

7. Advantages and Limitations

Advantages	Limitations
Reduces training time and memory footprint	PCA components are not interpretable in original feature terms
Mitigates curse of dimensionality	Information is irreversibly lost in lossy reduction methods
Feature selection improves model interpretability	Optimal feature subset selection is NP-hard (exponential search space)
Sampling reduces computation without losing statistical properties	Stratified sampling requires class labels — not applicable for unsupervised tasks
Removes noise features that cause overfitting	PCA requires standardisation first; sensitive to outliers in data

8. Case Study

CASE STUDY

Case: Face Recognition — Eigenfaces (PCA on Image Data)

Problem: A facial recognition system for security access control needed to process 64×64 pixel images (4,096 features per image) for 40 people × 10 images = 400 training images. Direct kNN on raw pixels was slow and inaccurate.

Dataset: AT&T Olivetti Faces dataset: 400 images, 4,096 features each.

Investigation: Applied PCA to the image matrix. Each principal component is an "eigenface" — a direction of maximum variance across all face images. The first few eigenfaces capture overall illumination, then facial structure, then fine details.

Scree Plot Analysis: Top 50 eigenfaces explain 93.7% of total variance. Top 150 explain 99.1%.

Result: Projected each image onto top-150 eigenfaces. kNN classification on 150-dim space:

- Original 4,096 features: Accuracy = 85.5%, Time per query = 12ms
- 150 PCA components: Accuracy = 94.5%, Time per query = 0.8ms

Conclusion: PCA reduced dimensionality from 4,096 to 150 (96.3% reduction) while IMPROVING accuracy by 9% and speeding up classification 15×. Noise features in raw pixels were harming the baseline model.

9. Scenario-Based Learning

SCENARIO 1

SCENARIO 1: PCA on High-Dimensional Genomics Dataset

Background: A bioinformatics team is building a cancer subtype classifier. Dataset: 500 patient samples, 20,000 gene expression features (microarray data). Training Random Forest takes 4 hours per run.

Technical Environment: Python, scikit-learn, pandas, 32GB RAM server.

Problem Statement: With 20,000 features and only 500 samples, the model severely overfits (train accuracy 99%, test accuracy 61%). Training time makes hyperparameter tuning impractical.

Available Evidence: $n=500$, $d=20,000$. $d \gg n$ is a classic high-dimensional, small-sample problem. RF variable importance shows >19,000 features with importance < 0.0001.

Investigation: Apply VarianceThreshold to remove near-constant genes, then PCA for dimensionality reduction.

Analysis: After VarianceThreshold, 15,200 genes remain. PCA scree plot shows 95% variance explained by 42 components. Reduce to 42 PCs.

Findings: RF on 42 PCs: test accuracy = 87%, training time = 8 minutes. Overfitting eliminated. Identified top 3 PCs correspond to known cancer subtypes (Luminal A, HER2+, Basal).

Recommended Solution: Apply VarianceThreshold first, then PCA with $n_components=0.95$. Validate using 5-fold cross-validation. Interpret top PCs against original gene loadings for biological insight.

SCENARIO 2

SCENARIO 2: LASSO Feature Selection for Credit Scoring

Background: A bank has 200 raw features for credit scoring: demographics, account history, transaction patterns, external credit bureau scores. Logistic Regression model with all 200 features has AUC=0.83 but is uninterpretable and slow.

Technical Environment: Python, sklearn LogisticRegression with L1 penalty, pandas, 500,000 loan records.

Problem Statement: Regulatory compliance requires the model to explain which features drive each decision. 200-feature black-box is rejected by the risk committee.

Available Evidence: 200 features. Many are highly correlated (e.g., 5 different credit score sources).

Pearson $r > 0.90$ for 23 feature pairs.

Investigation: Apply LASSO regularisation (L1 penalty) with cross-validated alpha selection. Plot feature coefficient path as alpha increases.

Analysis: At optimal $\alpha=0.01$, LASSO sets 173 of 200 coefficients to exactly zero. Retains 27 features including: payment_history_6m, debt_to_income, credit_utilisation, num_delinquencies_2y.

Findings: 27-feature LASSO model: AUC=0.82 (only 0.01 drop). Fully interpretable. Regulatory approval obtained. Inference speed: 3ms vs 18ms for 200-feature model.

Recommended Solution: Use LassoCV (cross-validated alpha) for automatic regularisation selection.

Report feature coefficients for compliance. Verify no protected attributes (race, gender) drive predictions.

SCENARIO 3

SCENARIO 3: Stratified Sampling for Imbalanced Fraud Dataset

Background: A payment processor has 10 million daily transactions with 0.1% fraud rate (10,000 fraud, 9,990,000 legitimate). Training any model on the full dataset is computationally expensive (6 hours per epoch).

Technical Environment: Python, sklearn, XGBoost, SMOTE library, 10M transaction records.

Problem Statement: Simple random sampling of 100,000 records gives only ~100 fraud cases — too few

for reliable fraud pattern learning. Model trained on random sample: fraud recall = 0.31.
 Available Evidence: Class distribution: Fraud 0.1%, Legitimate 99.9%. Random 1% sample: 100 fraud, 99,900 legitimate.
 Investigation: Apply stratified sampling to maintain 0.1% fraud rate in the sample, ensuring fraud class is proportionally represented.
 Analysis: Stratified sample of 100,000 records: 100 fraud (all available in 1% sample), 99,900 legitimate. Apply SMOTE on sampled data to synthetically oversample fraud class to 5,000 cases.
 Findings: Stratified sample + SMOTE: fraud recall = 0.79, precision = 0.71, training time = 8 minutes. Full dataset (6 hours): recall = 0.81, precision = 0.73. Negligible performance difference at 45× speed-up.
 Recommended Solution: Always use stratified sampling for imbalanced datasets. Combine with SMOTE or class weighting. Validate sample representativeness by comparing feature distributions between full data and sample.

10. Common Errors and Troubleshooting

Error	Cause	Impact	Fix
Applying PCA before standardisation	Skipping StandardScaler step	High-range features dominate PCs	Always StandardScale before PCA
Using PCA components for interpretation	Misunderstanding PCA output	PCs are linear combos — not original features	Use feature importance (RF/LASSO) for interpretable selection
Selecting k PCs from training set, applying to test	Correct	Standard practice is correct	Always fit PCA on training set only, transform test set
Simple random sampling on imbalanced data	Ignoring class distribution	Under-represents rare class (fraud, disease)	Use stratified sampling with class-proportional splits
RFE without cross-validation	Single train/test split for feature ranking	Overfits feature selection to training set	Use RFECV (cross-validated RFE) for stable feature ranking

11. Key Terminologies

Term	Definition
Dimensionality Reduction	Transforming data to fewer dimensions while preserving information (PCA, LDA, SVD)
PCA	Principal Component Analysis: unsupervised linear projection maximising variance
Explained Variance Ratio	Fraction of total variance explained by each principal component
Scree Plot	Bar chart of eigenvalues per PC; used to choose k (elbow point)
LDA	Linear Discriminant Analysis: supervised projection maximising class separability
SVD	Singular Value Decomposition: matrix factorisation $A = U \times S \times V^T$
Numerosity Reduction	Reducing number of data objects via sampling, clustering, or histograms

Stratified Sampling	Sampling proportionally from each class/stratum to preserve class distribution
Feature Selection	Choosing a subset of original features without transformation
Filter Method	Feature scoring independent of any ML model (chi-squared, mutual information)
Wrapper Method	Feature subset evaluated by ML model performance (RFE, forward selection)
Embedded Method	Feature selection during model training (LASSO, RF importance)
LASSO	L1-regularised regression that shrinks some feature weights to exactly zero
Curse of Dimensionality	Exponential data requirement as feature dimensionality increases

12. Topic Summary

UNIT II – Topic 5

DATA REDUCTION — SUMMARY

- Four strategies: Dimensionality Reduction (PCA, LDA, SVD), Numerosity Reduction (sampling, clustering), Data Compression (lossless, lossy, wavelet), Feature Subset Selection (filter, wrapper, embedded).
- PCA Steps: Standardise → Covariance matrix → Eigendecomposition → Sort by eigenvalue → Select k PCs (cumulative variance $\geq 90\%$) → Project.
- LDA: Supervised reduction maximising class separability. Max $k = \min(d, C-1)$.
- Feature Selection: Filter (chi-squared, MI — fast, model-agnostic), Wrapper (RFE — accurate, slow), Embedded (LASSO, RF importance — best trade-off).
- Sampling: Stratified sampling is essential for imbalanced datasets. Bootstrap for bagging/ensemble training.
- LASSO advantage: automatically sets irrelevant feature weights to exactly zero — built-in feature selection.
- Always: (1) standardise before PCA, (2) fit PCA/scalers on training set only, (3) use stratified sampling for class imbalance, (4) validate reduction quality with cross-validation.
- Tools: scikit-learn (PCA, LDA, TruncatedSVD, SelectKBest, RFE, VarianceThreshold, Lasso), UMAP, boruta, mlxtend.